nick mckenna

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Google Scholar: ↪ Published Work

Summary

- Applied Researcher at GitHub Copilot.
  Prev: Research Scientist at Microsoft Research.
- Publications in NLP/AI venues, including the *Best Paper Award* at AACL 2023.
- Areas of Expertise:
  - Designing agentic systems with LLMs & tools.
  - Training & contextualizing LLMs for code generation.
  - Solving natural language tasks, mitigating factual hallucination, & leveraging Knowledge Graphs.

Education

The University of Edinburgh

Ph.D. Informatics — 2023
Thesis: *Inference of Natural Language Predicates in the Open Domain*
Advisor: Mark Steedman

M.Sc. Artificial Intelligence, *Distinction* — 2019

Brown University

B.Sc. Computer Science — 2017

Professional Experience

Applied Researcher III, GitHub Applied Science — `London, UK, 2025.04 → Present`

Improving GitHub Copilot coding agents in cloud, CLI, and VSCode editor.

- Lead project to improve Copilot agent design with a new todo planning tool to create and manage work plans, improving performance metrics and expanding agent orchestration features.
- Created human annotation program for new internal evaluation benchmark.
- Analyzed competing agent designs to identify gaps in Copilot for VP-level project lead.

Research Scientist, Microsoft Research — `Cambridge, UK, 2024.02 → 2025.04`

Member of Calc Intelligence, the Excel AI research team.

- Lead project to finetune code-generating LLMs for low-resource programming languages (e.g. Excel Formulas) and APIs, useful for integrating code generating LLMs with native applications.
- Developed synthetic data generation techniques which were adopted by sister team.

Applied Scientist II Intern, Amazon, Alexa AI — `Cambridge, UK, 2022.06 → 2022.11`

- Developed a prototype for Alexa's trivia question-answering, and published the research findings. I improved the neural architecture to support knowledge graph expansion with no cost of retraining.

Skills

- **Modeling:** Designing coding agents with tools and computer access; Training code-generating models in multi-GPU environments; Generating synthetic data for finetuning models; Mitigating factual hallucination in LLM tasks by grounding to KBs and source text; Statistical analysis and classical ML.
- **Languages:** Python, TypeScript, Java, Bash, SQL/KQL
- **Tools:** PyTorch, Huggingface Transformers, LLMs like GPT-5 & Llama, Git, Numpy, Scipy

Patents, Awards, and Honors

US Patent (Pending) "Generating Interactions for Question Answering on Structured Data"	2025
The *Best Paper Award* at AACL 2023: *Smoothing Entailment Graphs with Language Models*	2023
Huawei Ph.D. Scholarship Award	2019
Outstanding M.Sc. Dissertation: *Learning Negation Scope Semantics with Structure*	2019

Scientific Service

Conference Reviewer

ARR (ACL) 2024 → Present
*ACL Conferences 2020 → 2023
ICLR 2024
COLING 2020

Talk Panelist

"Starting in NLP Research" NAACL, 2021

Teaching

Graduate TA at the University of Edinburgh

Accelerated NLP 2019 → 2021
NLU and Machine Translation 2020 → 2022

Undergraduate TA at Brown University

Computational Linguistics 2017
Computer Graphics 2016
Computer Architecture 2015

Research Publications

TBA 2026 — Mark Steedman, **Nick McKenna**, et al. *Combining Neural Language Models and Formal Methods for Natural Language Inference*. In Preparation.

TBA 2026 — **Nick McKenna**, Xinnuo Xu, Jack Williams, Nicholas Wilson, Benjamin Van Durme, Christian Poelitz. *Synthetic Function Demonstrations Improve Generation in Low-Resource Programming Languages*.
↪ Link

TBA 2026 — Christian Poelitz, **Nick McKenna**. *Synthetic Clarification and Correction Dialogues about Data-Centric Tasks: A Teacher-Student Approach*.
↪ Link

NeurIPS 2025 — Jinyang Li, Jack Williams, **Nick McKenna**, Arian Askari, Nicholas Wilson, Reynold Cheng. *Unlocking SLM Potential for Data Analysis Code Generation via Non-Parametric Knowledge Distillation*.
↪ Link

AAAI 2025 — Bhuvanashree Murugadoss, Christian Poelitz, Ian Drosos, Vu Le, **Nick McKenna**, Carina Suzana Negreanu, Chris Parnin, and Advait Sarkar. *Evaluating the Evaluator: Measuring LLMs' Adherence to Task Evaluation Instructions*.
↪ Link

EMNLP Findings 2023 — **Nick McKenna*** and Tianyi Li*; Liang Cheng, Mohammad Javad Hosseini, Mark Johnson, and Mark Steedman. *Sources of Hallucination by Large Language Models on Inference Tasks*.
↪ Link

AACL 2023 — **Nick McKenna**, Tianyi Li, Mark Johnson, and Mark Steedman. *Smoothing Entailment Graphs with Language Models*. ⋇Best Paper Award⋇
↪ Link

SustaiNLP 2023 — **Nick McKenna** and Priyanka Sen. *KGQA Without Retraining*. Workshop at ACL.
↪ Link

EMNLP 2021 — **Nick McKenna**, Liane Guillou, Mohammad Javad Hosseini, Sander Bijl de Vroe, Mark Johnson, and Mark Steedman. *Multivalent Entailment Graphs for Question Answering*.
↪ Link

CASE 2021 — Sander Bijl de Vroe* and Liane Guillou*; Miloš Stanojević, **Nick McKenna**, and Mark Steedman. *Modality and Negation in Event Extraction*. Workshop at ACL.
↪ Link

SEM* 2020 — **Nick McKenna and Mark Steedman. *Learning Negation Scope from Syntactic Structure*.
↪ Link